

Fusion of weighted Voronoi diagram and A* algorithm for mobile robot path planning

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Abstract—A* algorithm is one of the most common algorithms in path planning for mobile robots, which can complete global search and seek the lowest cost path. a* algorithm has more search ability compared with other algorithms, but its algorithm has long computation time, more search nodes and other drawbacks have always existed. To address these problems, this paper proposes a way to plan paths by fusing weighted Voronoi map and improved A* algorithm, which will adopt the concept of A* algorithm and combine with the idea of map construction, taking the lead to take the weighted Voronoi point as the priority expansion node, and if there is no directly searchable generation value minimum weighted Voronoi node, the weighted Voronoi edge will be the priority path, and so on. Simulation experimental results show that the algorithm can search for shorter paths in complex environments with relatively faster speed, which not only improves the search efficiency, but also improves the safety and accuracy of the paths.

Keywords—A* algorithm; Map modeling; Weighted Voronoi diagram

I. INTRODUCTION

Path planning is the most fundamental problem in robot navigation, which refers to how to plan a collision-free suitable trajectory from the departure position to the target position [1]. A variety of path planning algorithms have been studied by domestic and foreign scholars, such as A* algorithm, artificial potential field method, ant colony algorithm, etc. The traditional A* algorithm is widely used in mobile robot path planning, but the algorithm has problems such as a large number of inflection points and long search time when planning the path. In order to solve these problems, Chen et al proposed to combine the idea of artificial potential field with the traditional A* algorithm by assigning repulsive field functions to obstacles in the raster map and calculating the repulsive magnitude of the surrounding raster to improve the search capability of the A* algorithm[2]. Wu et al proposed an adaptive improved heuristic function to speed up the search

path of the A* algorithm, but the search path is long [3]. Wang et al proposed an improved A* algorithm based on obstacle prediction, which enabled the mobile robot to predict the obstacles in advance and search for the best paths otherwise[4], but the algorithm was computationally intensive and not efficient in operation.

Many scholars have improved the A* algorithm to make it improve compared with the traditional algorithm, but there are also long search time and large computation of the algorithm. In order to solve the above problems, this paper proposes an improved method of fusing weighted Voronoi diagram and A* algorithm, constructing a new simulation map consisting of weighted Voronoi edges and weighted Voronoi points as well as obstacle boundaries, and fusing raster diagram, with the help of A* algorithm's thinking path finding, and finally using Bessel curves to smooth the path turns to improve the traditional A* algorithm in terms of path length, path safety and search time.

II. ALGORITHM DESCRIPTION

A. Map Construction

To realize the global path planning of the mobile robot firstly, the map modeling of the operating environment is required, by collecting the obstacle environment, so the map modeling is the prerequisite to complete the path planning work, where the raster map method is the classical method of environment modeling, the raster map method cuts the map into raster grids of the same size and interconnected, when the mobile robot is in any raster position in the raster map, corresponding to the next moving position for the next eight adjacent grids of that grid [5]. By setting the obstacle value of each grid of the raster map to 0 and the no obstacle value to 1, and by dividing this workspace into $R \times S$ grids of the same size, if the information of all the grids in the 2D workspace is represented by Ω , we have:

$$\Omega = \sum N_{ij}, i \in [1, R], j \in [1, S] \quad (1)$$

In the formula (1): N_{ij} indicates the single grid information in the grid map: if $N_{ij}=0$, the grid corresponds to the free area; if $N_{ij}=1$, the grid corresponds to the obstacle area; R indicates the number of horizontal grids, S indicates the number of vertical grids.

The obstacle filling method is that the information of each grid corresponds to an area in the environment directly, and the filling method is that the corresponding area contains an obstacle is an obstacle grid, and the rest are free grids.

B. A* Algorithm

The A* algorithm is a heuristic search algorithm with a wide range of application scenarios, which is often used in path planning due to its high search efficiency and accuracy, and it mainly performs path planning by evaluating the current path points through a heuristic function. Its cost estimation function is:

$$F(n) = g(n) + h(n) \quad (2)$$

where: $g(n)$ is the cost function, which represents the actual cost from the starting node to the current node n ; $h(n)$ is the heuristic function, which represents the predicted cost from the current node n to the target point.

$$g(n) = \sqrt{(x_n - x_s)^2 + (y_n - y_s)^2} \quad (3)$$

$$h(n) = \sqrt{(x_n - x_t)^2 + (y_n - y_t)^2} \quad (4)$$

where: x_s is the horizontal coordinate of the starting point; y_s is the vertical coordinate of the starting point; x_n is the horizontal coordinate of the current node; y_n is the vertical coordinate of the current node; x_t is the horizontal coordinate of the target node; y_t is the vertical coordinate of the target node [6]. A* algorithm has stronger search capability and more accurate planning path compared with other global path search algorithms, but its calculation of this algorithm in complex environment long time and many search nodes drawback has been existed.

III. IMPROVED A* ALGORITHM FOR PATH PLANNING

A. Map modeling and pre-processing

Map modeling is very important for the path planning of mobile robots, and the model built can basically restore the environment of the real scene, so a good environment modeling method is the basis for completing the path planning. In this paper, we propose a map modeling method that integrates the weighted Voronoi diagram and the raster map method, taking into account the actual scene and the convenience of modeling.

1) Weighted Voronoi diagram.

The weighted Voronoi diagram is an extended form of the traditional Voronoi diagram, and its definition is similar to that of the traditional Voronoi diagram. Let P_i ($i=1, 2, \dots, n$) be n discrete points on a two-dimensional Euclidean space and λ_i

($i=1, 2, \dots, n$) be a given n positive real numbers, then it is said that:

$$V(P_i, \lambda_i) = \bigcap_{i \neq j} \left\{ P \mid \frac{d(P, P_i)}{\lambda_i} < \frac{d(P, P_j)}{\lambda_j} \right\} \quad (5)$$

is the Voronoi region of point P_i with weight λ_i , or called the weighted Voronoi diagram with P_i as the parent point [7]. In this paper, we will generate a weighted Voronoi diagram with the center of the outer circle of the obstacle as the occurring element, and the distances of the weighted Voronoi edges to the surrounding circles are all equal, as shown in Figure 1, where $w_1=u_1, w_2=u_2$.

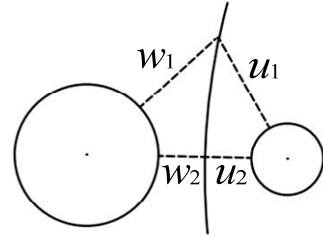
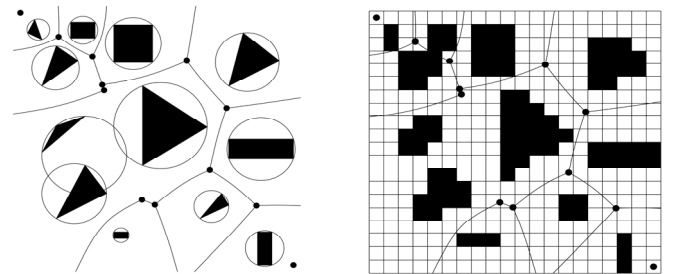


Figure 1 Partially weighted Voronoi diagram

2) Create rasterized weighted Voronoi diagrams.

On the basis of the existing obstacle information, the external circle of the obstacle is used as the regional boundary to get the weighted Voronoi edge of the adjacent obstacle, if $w_2 \leq 0$ then the weighted Voronoi edge of the two obstacles is not generated, and the point where the weighted Voronoi edge intersects two by two is the weighted Voronoi point. In order to avoid the occurrence of weighted Voronoi edges intersecting with the boundary of obstacles, the minimum distance z between weighted Voronoi edges and the outer circle of obstacles is set when generating the weighted Voronoi diagram to ensure the safety of the path generated later.

Randomly insert some triangles and rectangles as obstacles in the map, as shown in Figure 2(a), construct the weighted Voronoi diagram $V(P_i, \lambda_i)$ of obstacles, the generated weighted Voronoi edges form the set of path segments $L\{L_i \mid i=1, 2, \dots, n\}$, the set of weighted Voronoi points $M\{M_i \mid i=1, 2, \dots, n\}$, after the filling of obstacles, the maximum representation of each obstacle on the raster map, and retain the original weighted Voronoi diagram to get the improved raster map, as shown in Figure 2(b).



(a) Weighted Voronoi diagram (b) Obstacle rasterization

Figure 2 Model Map

B. Path Smoothing

The splicing path derived from the path planning algorithm still has the problem of more folds and larger corners, which will have a greater impact on the mobile robot's movement efficiency and wear, so the path needs to be smoothed. In this paper, we introduce Bessel curves to smooth the path of this algorithm.

Bessel curve is developed from Bernstein polynomial, which has been more widely used in industry because of its simple control, strong graphical description ability and smooth connection of position points. Assuming that $P(0)$ and $P(1)$ are the starting and ending points of the Bessel curve, respectively, and $P(u)$ denotes the control point of Bessel curve motion, the n th Bessel curve can be expressed as:

$$P(u) = \sum P(i)B_{i,n}(u), u \in (0,1) \quad (6)$$

where: u denotes the independent variable; $P(i)$ denotes the location point, corresponding to all location points $P(i)$ forming a characteristic polygon; $B_{i,n}(u)$ is the n th Bernstein polynomial [8], satisfying:

$$B_{i,n}(u) = C_n^i u^i (1-u)^{n-i} = \frac{n!}{(n-i)!i!} u^i (1-u)^{n-i} \quad (7)$$

where $i \in \{0,1,\dots,n\}$.

To ensure smooth steering at the nodes and to reduce the energy loss of the mobile robot during turns, the path turns will be smoothed using second-order Bessel curves.

C. Algorithm improvement and fusion

The Simple A* algorithm finds the optimal path by sequentially expanding the nodes with the smallest heuristic function. It starts from the starting node and keeps searching for nodes that reach the target point at the lowest cost, prioritizing the expansion of those nodes with smaller predicted values, thus forming a node set.

Firstly, the A* algorithm concept will be used, combined with the pre-raster map, to take the lead in expanding the weighted Voronoi point as the priority node, and if there is no directly searchable next weighted Voronoi node then the weighted Voronoi edge will be the priority path until reaching the next weighted Voronoi point, and so on, this method divides the nodes into weighted Voronoi points and alternative nodes, which can be selected and saved by direct search judgment, and the path diagram is drawn.

The steps are as follows.

Create an OpenList, a CloseList.

- Step 1: The weighted Voronoi diagram $V(P_i, \lambda_i)$ of obstacles is constructed, the generated weighted Voronoi edges form the set of path segments $L\{L_i | i=1,2,\dots,n\}$, the set of points $M\{M_i | i=1,2,\dots,n\}$ formed by the intersection of the weighted Voronoi edges, and the map model is built based on the raster method, the starting point d_s of the algorithm is set with the target point d_e , and the starting point is put into the OpenList.

- Step 2: Search the nodes to be searched in the set M with the parent node and determine if there is a child node in the set M that can be reached by direct reach or weighted Voronoi edge, if the point exists, repeat step 3. If it does not exist search the OpenList list to the child node with traditional A* algorithm. And save the path segment between the child node and the parent node.
- Step 3: Expand the search $M\{M_i | i=1,2,\dots,n\}$ from the parent node, and judge the relationship between the size of the minimum path cost $F(n)$ by the principle of A* algorithm, and take M_i with smaller path cost as the next search node, and judge whether the two-point straight line intersects with the Obstacle boundary set, if not, save the node to the CloseList, save the path segment of the two points, if it intersects, save the node to the CloseList, save the path segment of the two points.
- Step 4: And so on, after searching the node, determine whether it can intersect with the line composed of M_i in the CloseList and Obstacle boundary set, intersect then repeat step two, not intersect then update the CloseList, delete redundant nodes, repeat the operation.
- Step 5: The stitching process is performed for each node, and a second-order Bessel curve is introduced to smooth the algorithm stitching path.

IV. SIMULATION EXPERIMENT AND COMPARISON

In order to verify the effectiveness of the improved A* algorithm, a computer simulation of the improved A* algorithm was conducted. The simulation experimental environment is as follows: computer system environment is Windows 10, processor is Intel(R) Core(TM) i5-3470, running memory is 16GB, graphics processor is GTX 1050, and compilation environment is Matlab 2016b. The traditional A* algorithm and the improved fused A* algorithm are conducted within the same obstacle environment map. The comparison experiments were conducted. The simulated map is constructed with raster $N_{1,20}$ as the starting point and raster $N_{20,1}$ as the target point. The raster size is $20m \times 20m$, and the size of the raster is taken as $1m \times 1m$, and the minimum distance z is set as $0.1m$.

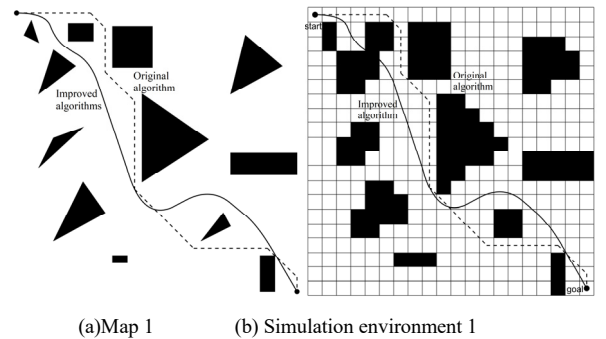


Figure 3 Two algorithmic paths under the map 1

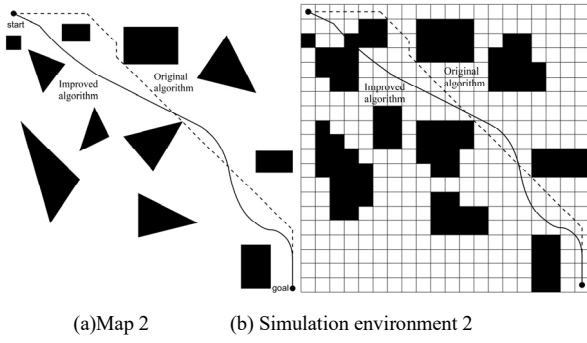


Figure 4 Two algorithmic paths under the map 2

From the operation maps shown in Figure 3 and Figure 4 can be seen A* algorithm in the planning of the path to search for the path within the eight neighborhood cost of the minimum principle, the search to the path can be relatively optimal, but in the complex obstacle environment will increase the search path node algorithm search time and path length, and will pass through the boundary points of the obstacle grid, will reduce the efficiency and accuracy of the search path, and the improved algorithm overall search node reduction, and the existence of applicable paths, reducing the search time, high path security, path length has also improved.

The results of the simulation experiments show that the improved A* algorithm searches for shorter path lengths, and that it expands fewer path nodes when searching for paths, thus reducing the time to search for paths. As can be seen from Table 1, in both environments, the path length planned by its improved A* algorithm compared to the traditional A* algorithm is shortened by 5.20% and 5.60%, respectively, and the number of nodes is significantly reduced by 79.31% and 58.33% on the path nodes, respectively, and the search time is reduced by 38.50% and 21.59%.

TABLE I. ALGORITHM PERFORMANCE METRICS COMPARISON TABLE

Map	Algorithm	Search time	Length	Nodes
Map1	Traditional	4.535	32.7279	29
	Improved	2.789	31.0264	6
Map2	Traditional	4.104	29.7990	24
	Improved	3.218	28.2179	10

Comprehensive evaluation shows that although the improved A* algorithm has less obvious advantages in path length, it reduces path nodes, shortens search time, and improves path smoothness and security.

V. CONCLUSION

In this paper, the A* algorithm is improved by integrating the weighted Voronoi diagram into the A* algorithm concept, and the following conclusions are drawn after conducting Matlab simulation tests. The improved algorithm can effectively reduce the threat points on the path and smooth out the path curve, which is suitable for the operation of mobile robots.

This algorithm is limited to the path planning of a single mobile robot in a static environment. In the future, the path planning of multiple mobile robots and the path planning of dynamic environments will be studied, so that the mobile robot can complete its tasks more quickly in different environments.

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